Rheumatoid arthritis (RA) is a chronic autoimmune disease in which the immune system attacks the synovium, the thin membrane lining the joints. The process runs in several overlapping stages.

1. Triggering the loss of tolerance

RA develops in genetically susceptible people exposed to environmental triggers.

- **Genetics:** The strongest risk factor is the HLA-DRB1 "shared epitope" alleles, which help present self-antigens to T cells. Variants in PTPN22 and other immune-regulatory genes add risk.
- **Environment:** Smoking is the best-established trigger. Periodontal infection (*Porphyromonas gingivalis*), gut and lung mucosal inflammation, and other infections are also implicated.
- **Citrullination:** Inflammation at mucosal sites (especially the lungs) activates PAD enzymes, which convert the amino acid arginine into citrulline in proteins. In people with the shared epitope, these altered proteins are seen as foreign. This is the origin of the anti-CCP antibodies (ACPA) found in most patients, often years before symptoms begin.

2. Adaptive immune activation

- Antigen-presenting cells show citrullinated peptides to CD4+ T cells (Th1 and Th17 types).
- Activated T cells help B cells, which produce **rheumatoid factor (RF)**, an antibody against the Fc portion of IgG, and **ACPA**.
- These autoantibodies form immune complexes that activate complement and engage Fc receptors on macrophages, amplifying inflammation.

3. Synovitis: the inflammatory cascade

Immune cells infiltrate the synovium, which becomes thickened and highly vascular.

- **Macrophages and synovial fibroblasts** release the key cytokines **TNF-α, IL-6, and IL-1**. TNF-α is a central driver, which is why anti-TNF drugs are so effective.
- **IL-17** (from Th17 cells) and **GM-CSF** further sustain inflammation.
- New blood vessel formation (angiogenesis) brings in more immune cells.
- Neutrophils accumulate in the synovial fluid and release enzymes and reactive oxygen species.

4. Pannus formation and joint destruction

The inflamed synovium becomes an invasive tissue called **pannus**, which erodes the joint.

- **Cartilage damage:** Activated synovial fibroblasts and chondrocytes release matrix metalloproteinases (MMPs) that degrade collagen and proteoglycans.
- **Bone erosion:** TNF-α, IL-6, IL-17, and RANKL promote differentiation of osteoclasts, which resorb bone at the joint margins. Suppression of Wnt signaling (via DKK-1) also reduces bone repair.
- Over time this causes joint deformity, instability, and loss of function.

5. Systemic effects

Circulating cytokines, especially IL-6, cause fatigue, anemia of chronic disease, and elevated CRP and ESR. They also raise cardiovascular risk and can cause osteoporosis. RA can affect the lungs, eyes, skin, and blood vessels.

6. Link to treatment

Therapy targets these mechanisms:

- **Methotrexate and other DMARDs** broadly suppress immune activation.
- **TNF inhibitors, IL-6 receptor blockers (tocilizumab), and CTLA4-Ig (abatacept)** block cytokines or T-cell co-stimulation.
- **Rituximab** depletes B cells.
- **JAK inhibitors** block intracellular signaling downstream of several cytokines.

In short, RA begins with a genetic and environmental breakdown of tolerance to citrullinated self-proteins. Autoreactive T and B cells then drive a cytokine-rich synovitis that develops into pannus, which destroys cartilage and bone.